

Prevention and removal of elevated radiocarbon contamination in the LLNL/CAMS natural radiocarbon sample preparation laboratory

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Abstract

The presence of elevated ^{14}C contamination in a laboratory preparing samples for natural radiocarbon analysis is detrimental to the laboratory workspace as well as the research being conducted. Elevated ^{14}C contamination occurs via many different pathways, but procedures can be adopted to prevent and remove this type of contamination. In this paper we present our current practices for preventing contamination, evaluating and monitoring potential workspaces and cleaning contaminated workspaces and laboratory equipment.

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1. Introduction

^{14}C is commonly used as a radiological tracer for biological and medical research and is now ubiquitous in the research environment. This widespread use of radiological ^{14}C poses a serious potential contamination problem for laboratories measuring natural levels of ^{14}C for conventional dating and other applications. Contamination can cost laboratory personnel and researchers a great deal of time, funding, stress and in severe cases lost research. For a multi-user facility such as ours,

there is also the possibility of cross contamination that may affect more than one user or set of samples. We address these issues and discuss the measures adopted at the LLNL/CAMS natural ^{14}C sample preparation laboratory to prevent and remove ^{14}C contamination.

2. Preventive measures within the sample preparation laboratory

Various measures can be taken within the sample preparation laboratory to prevent the introduction and later spread of contamination. Laboratory workspaces can be contaminated with ^{14}C as a result of earlier tracer based research, even if “hot” work has not occurred in the lab in

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decades. Elevated ^{14}C levels from earlier radiocarbon tracer based research can be detected in background samples decades later. The most common route of sample contamination is through contact with compromised surfaces. It is critically important that the user of any laboratory space make themselves aware of current and historical use of tracer levels of ^{14}C in the facility that houses their laboratory. If radiocarbon is currently being used within the building or in a nearby location, both pedestrian traffic and airflow can potentially contaminate the laboratory. It is essential that the laboratory not be a shared facility or at the very least that no common areas are used for radiocarbon tracer research.

Ventilation systems within such a facility should be segregated if possible and foot traffic adjacent to the laboratory should be restricted. Controlling access to the workspace itself is vital. Persons from hot labs should not enter the natural labs and vice versa. Installing locks and mounting signs that restrict entrance to the lab can help deter non-essential personnel.

Support personnel who have job responsibilities in more than one building such as maintenance, administrative and delivery personnel must be made aware of the potential for contamination by foot traffic. Maintenance activities and deliveries should be scheduled as to avoid possible contamination. For example, bi-weekly dry ice deliveries are made to the CAMS graphite prep lab prior to deliveries to radiological laboratories at the LLNL. The laboratory should also have its own tools and janitorial supplies.

Elevated ^{14}C can also be spread from collaborators' samples, equipment and personnel. Workspace surfaces (lab, research vessel, or field site) are assayed for contamination before commencing work in a new collaborator facility and periodically in facilities with ongoing projects. Any surface that has been used frequently by many people has the potential to be contaminated. To prevent legacy contamination we insist researchers who wish to submit samples to our laboratory test their labs, research vessels and field sites for ^{14}C contamination prior to submitting research samples. Surfaces routinely touched by many users, such as door-knobs, light switches, drawer handles, telephone

receivers, computer keyboards, faucet handles, lab stools, armrests, balances, apparatus controls, centrifuges, sonicators, and workbench tops should be assayed by taking swipes. In most cases, a workspace can be evaluated with 10 swipes.

A discussion of the procedure for obtaining swipes and interpreting the results is in the literature [1]. A small glass fiber filter (e.g. Whatman GF/A 21–24 mm) is wetted with isopropyl alcohol and wiped across the laboratory surface in question [1]. The swipe will pick up organic and inorganic material. A blank filter should be included in any set of swipes. The swipes should be individually packaged and sent to a bio-AMS sample preparation laboratory for conversion to graphite [2,3] and measured for ^{14}C content by AMS. It is not recommended that an AMS lab which prepares samples of natural ^{14}C levels process its own swipes. We process our swipes at the bio-AMS prep lab about 1 km downwind from our natural radiocarbon prep lab. Swipes are analyzed using our compact bio-AMS system [4].

The introduction of used equipment into the laboratory is not recommended. If second-hand equipment is used, it should be swiped, and the swipes measured before the equipment is brought into the building. Swipes of shipping containers or sample collection equipment can also be taken if there is any question concerning ^{14}C contamination. Since each swipe cost $\sim \$100$ to produce and analyze, inexpensive laboratory supplies should be purchased new from suppliers and not salvaged from other labs.

Workstation design can greatly influence laboratory cleanliness. Adhesive peel-away sticky mats should be placed at each doorway to make personnel aware of the potential for contaminants dispersed via "foot traffic". Within the lab, counter tops and workbenches should be covered in aluminum foil and re-foiled routinely in order to provide for clean sample processing areas. This practice also keeps the original underlying surface free of contamination. Using disposable glass/plastic ware (pipets, bottles, etc) is highly recommended to maintain a clean lab. Avoid obtaining disposable supplies from "on campus" stockrooms that have foot traffic from a large user base. We have received samples collected at remote field-

work sites that were contaminated from plastic sample bags obtained from the chemical store-room at a university. Reusable glassware such as beakers, flasks and watch glasses are washed in detergent, dried and baked at 400 °C for 1.5 h.

Equipment such as vacuum ovens, centrifuges, refrigerators and freezers are often extremely susceptible to contamination. Maintaining logbooks for all sample processes and equipment can help locate areas that have been contaminated and aids in the prevention of further contamination. This information functions as a vital tool for potentially identifying the source of contamination. Logbooks also help determine what equipment requires cleaning or disposal and which samples may be compromised after a contamination event by documenting the sample pathways through the lab. Long term sample archiving within the proximity of the lab is not recommended. If storage is a necessity, then separate, designated, clean storage areas for samples are recommended.

3. Responding to a contamination event

In one of our worst contamination events, a collaborator with a clean history and a lab in a new building sent samples that were probably scintillation countable. The collaborator's lab did not tightly control access and another researcher at his facility decided to use his high-speed centrifuge with ^{14}C -labelled material when the lab was vacant. The unauthorized user completely contaminated the lab. When we received their new samples, we assumed they were clean, processed them, contaminated our prep lab and destroyed an ionizer on our spectrometer. Tighter access control at our collaborator's lab would have prevented the episode. Had periodic surveillance swipes been taken at the collaborator's lab, we would have found the contamination. In nearly every case of contamination in our lab, a screening procedure was omitted or sidestepped.

Once contamination is detected in any lab, a significant effort is required to control its spread. In many instances, elevated ^{14}C is not recognized until both the unprocessed/unmeasured samples and the laboratory have been compromised. Prior

to detection of contamination, exchange of equipment and access of personnel allows the contamination to spread and potentially affect other samples, laboratories and research within the facility.

The initial response to a contamination event is to control the spread of the contaminant. This is accomplished by, suspending all sample processing and limiting laboratory access to mandatory personnel. Contaminated samples cannot be cleaned and are considered lost. The next step is to begin cleaning the workspace and supplies. Dispose of all replaceable suspect supplies (i.e. bottles, tubes, surface covers). Dispose of all laboratory chemicals that were used in the processing and pre-treating of the suspect samples. Thoroughly clean all laboratory surfaces (e.g. tables, floors, light switches, door handles, telephones, computer keyboards and other items handled or accessed by people). Recommended cleaning agents are detergent, bleach and ethanol. It is also important to clean and track down equipment exchanged with other laboratories or offices.

Graphitization rigs, vacuum manifolds and other equipment used to process samples will require dismantling and cleaning. Reduction rigs made of stainless steel, or glass need to be disassembled, washed in detergent, washed in a solvent if possible, rinsed with de-ionized water several times and dried in a convection oven.

Vacuum driers also require cleaning. All parts need to be wiped down with acetone and/or ethanol. Dispose of and replace all plastic tubes, gaskets, bearings and vacuum pump oil. Thaw the cold traps, remove the chemicals and clean the basin with acetone and/or ethanol. It should be noted that vacuum dryers and vacuum ovens are particularly difficult to clean due to the deposition of hot material all over the surfaces and then re-suspension and back flow under vacuum.

All tools and containers that come into contact with the samples should be properly disposed or cleaned. Glassware should be cleaned with acetone and then baked at the highest possible temperature. Tools used for loading samples should be cleaned in a manner similar to the manifold and graphitization rig parts. Clean heating blocks with acetone. It may even be necessary to bead blast the

surface of the blocks. Bake or dispose of sample catalysts and other additives (i.e. CuO, Ag).

Being contaminated by a sample that is less than 10 times modern warrants the same cleaning response as by a sample contaminated at greater than 10 times modern. Although a sample might be less than 10 times modern we have no real way of knowing if the sample was in fact “hotter” prior to being pre-treated in our lab. We have found it is in our best interest to always clean the lab as if it we were hit with ^{14}C contamination greater than 10 times modern. The only major difference in reactions to contamination is disposal of some susceptible components in known severe events. Severe contamination events often irreparably contaminate pressure transducers and O-rings, so they are simply replaced. With careful handling, it is possible to process a sample up to two times modern without contaminating the laboratory or reduction line.

The fixed costs of a contamination event are tied to the disruption of sample production and the effort required to clean, roughly \$15 000 per day at our facility. Total costs depend on the value of the samples lost. The high cost of lab shutdown is an incentive to successfully clean the lab on the first try.

By running background samples consisting of ^{14}C free material through the different stages of sample preparation we are able to determine if the cleaning was successful. Background samples work best if they have a large surface area and are highly absorbent: some possible background materials include; old wood, coal and petroleum products. It is important to test as comprehensively as possible. Samples should pass through all sample-handling equipment (centrifuges, ovens, heating blocks, weighing tables, scales and vacuum manifolds). Limit the exposure of individual background samples so that suspect samples can be tracked to a single piece of equipment or process. Continue to clean and run background samples until the laboratory is free of contamination. This process, if done properly, will rarely warrant repeating. Other laboratories or offices that exchange materials or equipment with the contaminated laboratory may also need to be tested with swipes as well as aerosol monitors.

Aerosol monitors consists of sorbent carbon (e.g. fullerene soot such as Alfa Aesar product #40971) with high surface area and depleted

^{14}C cations and may be used to detect airborne ^{14}C contamination. The soot is mixed with iron powder (for improved conductivity in the ion source) and packed directly into AMS targets, loosely wrapped in aluminum foil to protect against the introduction of dust particles, and left in the laboratory for 1–2 weeks. We have seen that volatile organic compounds are able to diffuse through the aluminum foil and are adsorbed on the soot. Samples are analyzed for ^{14}C using AMS [1]. Typical samples are slightly elevated at the surface due to the sorption of contemporary CO_2 and quickly drop off to background with more extended sputtering. Monitors in a constant hot air space retain elevated levels deep into the soot sample. Sample prep for aerosol monitors is minimal so costs are dependent on machine time, ~\$70 each on our compact AMS system [4].

4. Conclusions

The procedures discussed within this paper cover a number of different aspects dealing with the prevention and cleanup of contamination but are not all encompassing. The extrapolation of the procedures discussed can address issues for different laboratory arrangements and procedures. It is crucial that laboratory personnel and collaborators understand the threat of ^{14}C contamination. Collaborators who submit samples are responsible for ensuring that their samples are uncontaminated. Laboratory personnel are responsible for ensuring that the natural radiocarbon sample preparation laboratory remains clean.

^{14}C contamination cannot always be avoided, but if these procedures are executed properly the spread of contamination can be controlled and the laboratory restored. Omitting any of these precautions increases the likelihood of a contamination event.

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